

Naici Guo

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Empirical researcher with strong statistical background and domain expertise in healthcare and biomedical contexts. Specialised in hypothesis-driven data analysis, machine learning, and quantitative modelling using Python, R and SAS. Adept at translating complex data into actionable insights.

EXPERIENCE

University of Sheffield

Research Associate

May 2024 - Current

- Develop statistical pipelines in **Python** for genomics and real-world data analysis in **high-performance computing (HPC)** environments, using bioinformatics tools, Bayesian modeling method etc.

University of St Andrews

Teaching Assistant & Workshop Demonstrator

Sep 2019 - Nov 2023

- Led over **50+ tutorials and computing sessions** for undergraduate students, modules including Statistics in Practice, Python Computing Projects, Multivariate Calculus, and Mathematical Computing.

Accenture

Consulting Internship

May 2022 - Sep 2022

- Conducted extensive **evidence-driven research** across industries, including insurance, healthcare, and alternative fuel vehicles.
- Composed a **25-page analytical report** synthesizing research insights and data trends for senior stakeholders.

Parexel

Drug Safety Associate

Feb 2019 - Aug 2019

- Analysed **real-world healthcare data (EHR, drug safety reports)** to identify adverse drug reactions and treatment effects.
- Generated and reviewed **300+ pharmacovigilance reports per month**, ensuring compliance with regulatory standards.

EDUCATION

University of St Andrews

PhD in Statistics

St Andrews, UK

2019–2024

University of Birmingham

MSc in Toxicology, Merit

Birmingham, UK

2017–2018

Shenyang Pharmaceutical University

BSc in Pharmacy

Shenyang, CN

2012–2017

TECHNICAL SKILLS

- **Programming:** *Python, R, SAS, Bash script*
- **Statistical & Quantitative Modelling:** *Classification, Regression, Bayesian Inference, Survival Analysis, etc.*
- **Tools & Environments:** *JupyterLab, R Markdown, Git, Linux command line, Slurm & HPC etc.*
- **Reporting:** *Data visualisation, Technical writing, etc.*

PUBLICATIONS & PRESENTATIONS

1. **Guo, N.**, Minas, G., Synowsky, S.A., Dunne, M.R., Ahmed, H., McShane, R., Bhardwaj, A., Donlon, N.E., Lorton, C., O'Sullivan, J. and Reynolds, J.V., 2022. Identification of plasma proteins associated with oesophageal cancer chemotherapeutic treatment outcomes using SWATH-MS. *Journal of Proteomics*, 266, p.104684.
 2. Li, L., Chen, J., Qin, G., Qi, Y., Chen, Y., Li, M., Zhang, Q., Cheng, Y., **Guo, N.**, Moutari, S. and Moore, J.E., 2023. Tear Film Lipid Layer Changes Following Combined Effect of Heated Eye Mask with Intense Pulsed Light Therapy for Evaporative Dry Eye: A Randomized Control Study. *Photobiomodulation, Photomedicine, and Laser Surgery*, 41(8), pp.435-444.
 3. Chen, J., Qin, G., Li, L., Qi, Y., Xia, Y., Zhang, Q., Wu, Y., You, Y., Yang, L., **Guo, N.** and Moutari, S., 2023. The combined impact of intense pulsed light combined and 3% diquafosol ophthalmic solution on evaporative dry eye: a randomized control study. *Ophthalmology and Therapy*, pp.1-13.
 4. Qin, G., Chen, J., Li, L., Qi, Y., Chen, Y., Zhang, Q., Wu, Y., You, Y., Yang, L., **Guo, N.** and Moutari, S., 2023. Effects of diquafosol sodium ophthalmic solution on tear film matrix metalloproteinase-9 and corneal nerve density in patients with type 2 diabetic dry eye. *Journal of Ocular Pharmacology and Therapeutics*.
- 2023 **Maths Division Presentation:** "Evaluation the performance and effects of Immunoaffinity-based Columns for Human Plasma Depletion to Facilitate SWATH Proteomics"
 - 2022 **Maths Division Presentation:** "Analysis of Omics Data in Cancers"
 - 2021 **StAMBio seminar** "A Multivariate Model-based Approach for Treatment Response Prediction in Oesophageal Adenocarcinoma (OAC)"
 - 2019 **SMMB** "Comparison of Clustering/Classification Methods"
 - 2018 **Poster** "The effects of three cardioplegia solutions on myocardial metabolism during ischaemia-reperfusion in a Langendorff murine model"

GROUP PROJECTS & WORKSHOPS

ICMS and MAC-MIGS Modelling Camp

May 2022

Project: Mathematical modelling of carbon capture by adsorption

- Collaborated with other five fellows to analyse the mathematical model of the CO₂ adsorption process; solved and presented both linear and non-linear analytical solutions.
- Contributed by conducting simulation studies to test the effects of variables, data visualisation and presenting.

Correlation One

Aug 2020 - Oct 2020

Data Science for All/Women Summit

- Conducted a model-based feature importance analysis on survey data recognised the three most important social-economic factors that affect people's opinions on the COVID-19 vaccine.
- Guided the team with missing data imputation and normalisation for data from 26,000 surveys using Python, guaranteed the successful implementation of machine learning algorithms.